

MOHAMMAD HOSSEIN MOHAMMADI

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Research Interests

- Large Language Models (LLMs) & Generative AI
- Computer Vision & Medical Image Analysis
- Human-Centered AI & Intelligent Systems Design
- Explainable AI (XAI) & Trustworthy Machine Learning

Education

Islamic Azad University, Najafabad Branch

Sep. 2015 – Sep. 2021

Bachelor of Science in Computer Engineering (Information Technology)

Najafabad, Iran

- **Thesis Title:** Semantic segmentation of breast cancer pathology images using the U-Net model. 
- **Supervisor:** [Dr. Mehdi Jabalameli](#)

Research Experience

Research Assistant | *Sup: Dr. Seyed Zachariah Moradi & Aram Salehi*

2026 – Present

- Currently co-authoring a research paper titled “Artificial Intelligence in Neurodegenerative Disease Diagnosis.”

Research Assistant | *Machine Learning Laboratory, Isfahan University | Sup: Dr. Amirhassan Monajemi*

2018 – 2019

- Conducted research on advanced concepts in Computer Vision, Machine Learning, and Signal and Image Processing.

Technical Speaker | *Open Source Software Conference & Ubuntu 17.04 Release Party*

Jul 2017

- Delivered a presentation on “How to Prevent DOS & DDOS Attacks on Linux Servers,” sharing practical network security knowledge with the open-source community.

Executive & Technical Program Committee Member | *Int'l Conference on CS & Engineering*

Oct 2016 – Apr 2017

- Collaborated with academic staff to organize technical panels, manage event logistics, and evaluate technical programs to ensure the successful execution of the conference.

Publications

Patents

- [1] **The system for early detection of Heterochromia disease in the neonates**  2021
Seyed Yahya Moradi, M.H. Mohammadi | Patent No: IR-139850140003011312
- [2] **Pocket Protector Of Sensitive Injective And Inhalant Vials Against Temperature And Impact Changes**  2020
Seyed Yahya Moradi, M.H. Mohammadi, Pejman Mohammadi | Patent No: IR-139850140003011419
- [3] **Energy Waste Reduction System based on prediction of people's presence and working time in industries and medical centers**  20..
S.Y. Moradi, M.H. Mohammadi, A. Anvari | Patent No: IR-139850140003011094 | Iran Patent
- [4] **Coupling device temperature to body temperature for ambulances and hospitals sera**  20..
S.Y. Moradi, S.Z. Moradi, M.H. Mohammadi | Patent No: IR-139550140003013173 | Iran Patent

Articles

- [J1] **Computer Modeling of Information Transmission Principles and Methods in Telemedicine Networks**  2018
S.Y. Moradi, M. Akbari Moghaddam, M.H. Mohammadi | Journal of Rafsanjan University of Medical Sciences (JRUMS)
- [C1] **Telemedicine Methods; Basics, Design, and Optimization**  2018
S.Y. Moradi, M. Akbari Moghaddam, M.H. Mohammadi | 3rd International conference on Modern Approaches in Science, Technology and Engineering

Books

- [1] **A Step-By-Step Guide to Patenting in Iran**  2020
M.H. Mohammadi, Ali Karimi, S.Y. Moradi | Publisher: Arna | ISBN: 978-600-356-922-8
- [2] **Basics of Internet of Things (IoT)**  2021
S.Y. Moradi, M.H. Mohammadi | Publisher: Arna | ISBN: 978-622-2910-14-3
- [3] **Principles Of Image Compression in The Medical and Aerial Image By MATLAB Software**  2021
A. Salehi, S.Y. Moradi, M.H. Mohammadi, M. Radaei | Publisher: Arna | ISBN: 978-622-291-001-3

Projects

- Multimodal AI: Image Caption Generation System** | *Python, CNN, LSTM, NLP, Computer Vision* |  **May 2026**
- Developed an end-to-end image captioning system using a hybrid CNN-LSTM architecture trained on the Flickr8k dataset to generate contextual descriptions.
- End-to-End Speech Recognition Web Service** | *Python, Deep Learning, Streamlit, Git, UI/UX* |  **Sep 2025**
- Developed and deployed an end-to-end deep learning Speech Recognition system with an interactive Streamlit web interface for real-time audio processing and transcription.
- HealthTech AI: Medical Image Segmentation** | *PyTorch, U-Net, FastAPI, Docker, UI/UX* |  **Jan 2025**
- Developed a Dockerized breast cancer image segmentation system using U-Net and FastAPI, providing an intuitive web interface for clinical analysis.
- Reverse Engineering: Video Borescope Device** | *Hardware Design, Prototyping, Reverse Engineering* |  **Oct 2017**
- Co-designed and reverse-engineered industrial video borescopes featuring interchangeable probes (2.4mm–8.4mm) and 4-way joystick articulation for advanced machinery endoscopy.

Work Experience

- Product Designer (Design System Specialist)** **Sep. 2025 – Present**
TeamNey *Scotland, United Kingdom (Remote)*
- Architected a robust Design System with unified design tokens to accelerate frontend development, while bridging design-engineering gaps to ensure accessible, pixel-perfect implementations in a remote environment.
- UI/UX Designer** **Feb. 2022 – Sep. 2023**
PizzaBar *Copenhagen, Denmark*
- Led the end-to-end design of a custom CRM by developing a scalable Figma design system, conducting usability testing, and transforming user research into interactive prototypes to improve workflows.

Honors & Awards

- 2nd Place Winner, 7th Mobile Programming Marathon** | *Sharif University of Technology* |  **Sep. 2019**
- 7th Mobile Programming Marathon along with the "Chelesme" team members, 85 teams participated from Iran, Sharif University of Technology.
- 3rd Place Winner, 4th Mobile Programming Marathon** | *Sharif University of Technology* |  **Sep. 2017**
- 4th Mobile Programming Marathon along with the "Yakhmak" team members, 90 teams participated from Iran, Sharif University of Technology.

Skills

Languages & Computing: Python, C++, Java, JavaScript, SQL, NumPy, Pandas
LLMs & Generative AI: Transformers, LLM Fine-Tuning, RAG, LangChain, Hugging Face, Prompt Engineering, Diffusion Models
Computer Vision & DL: PyTorch, OpenCV, Medical Image Analysis, Image Segmentation (U-Net), ViT, XAI (SHAP, LIME)
AI Infrastructure & MLOps: Docker, FastAPI, MLflow, Git, GitHub Actions CI/CD, AWS
HCI & Intelligent Systems: Figma, Design Systems, Streamlit, Interactive Prototyping, Human-in-the-Loop AI

Languages

- English (IELTS: 7.0 / S:7, W:7, R:7, L:7)
- Persian (Native proficiency)

Professional References

References available upon request.